



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI
MIDTERM EXAMINATION OF BS ARTIFICIAL INTELLIGENCE
2ND SEMESTER 1ST YEAR BATCH 25F SECTION (A,B,C)

COMMUNICATION SKILLS (THEORY)

DATED: 30-03-2026 TIME ALLOWED: 45 MINUTES (12:00pm) MAX. MARKS 15

Student Name : _____

Student Id Number : _____

Instructions:

1. This paper contains 02 questions.
2. Use of Calculator is not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be canceled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationary is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned back to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q # 01 (a) Define Communication Skills, and elaborate on their importance.
[C-2, CLO-01, PLO-09, 04 Marks]

Q # 01 (b) Compare and contrast verbal and non-verbal communication with suitable examples.
[C-2, CLO-01, PLO-09, 04 Marks]

Q # 02 Explain the 7 C's of effective communication and their role in making communication effective.
[C-2, CLO-01, PLO-09, 07 Marks]

THE END

MIDTERM EXAMINATION 2026 OF 2nd SEMESTER, 1st YEAR (2025F BATCH)
OF B.S (ARTIFICIAL INTELLIGENCE) SECTIONS (A1, A2 & A3)

DIGITAL LOGIC DESIGN (THEORY)

DATED 26-03-2026

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name : _____

Student Id Number : _____

Instructions:

1. This paper contains 02 questions.
2. Use of Calculator is allowed
3. Cheating of any type will disqualify the candidate.
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NOTE; - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.No.	Question	CLO Assessed	Marks																								
01	(a) Describe a timing diagram. You are given three input signals A, B, and C with the following timing behavior: <table border="1"> <thead> <tr> <th>Time interval</th><th>A</th><th>B</th><th>C</th></tr> </thead> <tbody> <tr> <td>$t_0 - t_1$</td><td>0</td><td>0</td><td>1</td></tr> <tr> <td>$t_1 - t_2$</td><td>1</td><td>0</td><td>1</td></tr> <tr> <td>$t_2 - t_3$</td><td>1</td><td>1</td><td>0</td></tr> <tr> <td>$t_3 - t_4$</td><td>0</td><td>1</td><td>0</td></tr> <tr> <td>$t_4 - t_5$</td><td>1</td><td>1</td><td>1</td></tr> </tbody> </table> Logic : $Y = (A \cdot B + \bar{C})$ (i) Draw the timing waveforms for inputs A, B, and C. (1) (ii) Determine and draw the intermediate signals: (1) • $A \cdot B$ (AND output) • $\bar{C}$ (NOT output) (iii) Using the above, draw the final output waveform Y. (1) (iv) Clearly mark transitions at each time interval. (1) (b) Convert $(7F)_{16}$ to binary. (1) (c) Convert $(345)_8$ to hexadecimal. (1) (d) Draw a truth table of 3 input AND gate & Express the Boolean Expression. (2)	Time interval	A	B	C	$t_0 - t_1$	0	0	1	$t_1 - t_2$	1	0	1	$t_2 - t_3$	1	1	0	$t_3 - t_4$	0	1	0	$t_4 - t_5$	1	1	1	CLO-1	8
Time interval	A	B	C																								
$t_0 - t_1$	0	0	1																								
$t_1 - t_2$	1	0	1																								
$t_2 - t_3$	1	1	0																								
$t_3 - t_4$	0	1	0																								
$t_4 - t_5$	1	1	1																								
02	(a) Simplify following boolean expressions using boolean algebra. (i) $(A \cdot B) + (\bar{A} + B) + (A \cdot \bar{B})$ (2) (ii) $((A + B)(\bar{A} + B))$ (1) (iii) $\bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C + A\bar{B}C + ABC$ (2) (iv) $[AB(C + \bar{B}D) + \bar{A}B]CD$ (2)	CLO-2	7																								

THE END



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

MIDSEMESTER EXAMINATION OF 2ND SEMESTER 1ST YEAR OF BATCH-2025/FALL
ARTIFICIAL INTELLIGENCE OF 2026

SUBJECT: FEHME QURAN (THEORY)

DATED: 28-03-2026

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 00 questions.
2. Use of Calculator is allowed/Not allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q 1 : Related to words mostly used in Urdu language. (Marks 06)

Q 1 (A)	CLO 01	Marks (02)
What do you know about the following words? Provide a brief explanation for each.		درج ذیل الفاظ کے بارے میں آپ کیا جانتے ہیں؟ ہر ایک کو مختصر وضاحت کریں۔

No.	Arabic Word / لفظ	(Urdu/English)
1	الْقُرْآن	
2	الْكَفَّة	
3	الْجَنَّة	
4	الْخَيْفَان	
5	مُؤَسَّس	
6	الْوَشْرُك	
7	الرَّبِّ	
8	الْرَسُول	

Q 1 (B)	CLO 01	Marks (01)
Complete the following incomplete words by filling in the blanks.		درجہ بالا الفاظ کو مکمل کریں۔

No.	Incomplete Arabic Word / نامکمل لفظ		Incomplete Arabic Word / نامکمل لفظ
1	الْحَلَنَ _____	6	الْخَ _____
2	الْمَخْرَا _____	7	الْمَنْطَا _____
3	الْفَسَا _____	8	الْمُطَوَا _____
4	الْمُسَوَا _____	9	الْمُسَرَا _____
5	الْمُزَعَا _____	10	الْمُزَا _____

Q 1 (C)	CLO 01	Marks (01)
Translate the following words into Arabic.		درجہ بالا الفاظ کو عربی میں لکھیں؟

Meaning/معنی	Word/لفظ		Meaning/معنی	Word/لفظ	
	مشرق	6		کلام	1
	مغرب	7		منافع	2
	طلوع	8		عدل	3
	رزق	9		عزت	4
	مجاہد	10		ذلت	5

Q 1 (D)	CLO 01	Marks (02)
Apply the appropriate diacritical marks (Zair, Zabar, Paish, Jazm, Shad/ Tashdeed) to the following words.		نیچ دی گئے الفاظ پر مناسب حرکات (زیر، زبر، پیش، جزم، شد/ تشدید) لگائیں۔

الْخَلِيفَةُ	الْجُمُعَةُ	النَّهْرُ	الْقُبْلَةُ
التَّوْفِيقُ	الْإِيمَانُ	الْبَحْرُ	الْعَرَبِي

Q 2: : Related words which are generally comprehensible with little effort for speakers of Urdu, Sindhi, Pashto, Persian, and Balochi. (Marks 04)

Provide the Urdu or English meanings for the following Arabic words in the space provided.	درج ذیل عربی الفاظ کے اردو یا انگریزی معنی دی گئی جگہ پر لکھیں۔
--	---

معنی 'ی	لفظ / Word		معنی 'ی	لفظ / Word	
	الْقَلِيلُ	5		الْبَيْتُ	1
	الْإِسْرَافُ	6		الْجَمَالُ	2
	الْعَاقِبَةُ	7		النَّسَبُ	3
	الْمَوَدَّةُ	8		الْشَوْبَةُ	4

Q3: Related to Quranic Word Formation. (Marks 05)

Q 3 (A)	CLO 01	Marks (= 2.5)
Write the singular forms (واحد) of the following words:		درج ذیل الفاظ کے واحد لکھیں۔

Singular/واحد	Plural/جمع		Singular/واحد	Plural/جمع	
---------------	------------	--	---------------	------------	--

1	آنهار	6	أموال
2	رسل	7	أعمال
3	أيام	8	أشياء
4	فبور	9	أسماء
5	قلوب	10	خُدود

Q 3 (B)	CLO 01	Marks (2.5)
Write Urdu / English meaning of the following short sentences:		درج ذیل مختصر جملوں کا مطلب / مفہوم لکھیں۔

قرآن مجملہ	مفہوم	Meaning	قرآن مجملہ	مفہوم	Meaning
1	هَذَا بَلْع	6	هَذِهِ ال نُنْيا	مفہوم	Meaning
2	هَذَا هَذِ ی	7	هَذِهِ الْبَلْدَة	مفہوم	Meaning
3	هَذَا عَدُوْ	8	هَذِهِ ال شَجَرَة	مفہوم	Meaning
4	هَذَا نَذِيْر	9	هَذِهِ الْحَيَاة	مفہوم	Meaning
5	هَذَا فَوْج	10	هَذِهِ النَّار	مفہوم	Meaning

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI
MIDSEMESTER EXAMINATION OF 2ND SEMESTER 1ST YEAR OF BATCH.
2025/FALL BS ARTIFICIAL INTELLIGENCE SEC (A, B & C) OF 2026

SUBJECT: LINEAR ALGEBRA (THEORY)

DATED: 24-03-2026

TIME ALLOWED 90 MINUTES

MAX. MARKS 30

Student Id Number: _____

Student Name: _____

Instructions:

1. This paper contains 04 questions.
2. Use of Calculator is allowed/Not allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO.1(a) Describe:

- Scalar matrix
- Diagonal matrix
- Skew Symmetric matrix

Q.NO.1(b) Interpret the rank of matrix

$$\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ 3 & 1 & -2 \end{bmatrix}$$

(CLO-1, GA-2, C2, 7.5 MARKS)

Q.NO.2. Solve the following system of equations by using Gauss elimination method

$$3x + 4y + 5z = 40$$

$$2x - 3y + 4z = 13$$

$$x + y + z = 9$$

(CLO-2, GA-3, C3, 7.5 MARKS)

Q.NO.3. Execute the following system of equation using Cramer's Rule

$$x + 2y + 3z = 14$$

$$2x + 3y + z = 13$$

$$3x + y + 2z = 13$$

(CLO-2, GA-3, C3, 7.5 MARKS)

Q.NO.4. Illustrate the Inverse of Matrix and Interpret the system of linear equation by using matrix method

$$2x - 3y + 4z = 6$$

$$3x + 4y - 5z = 7$$

$$4x - 5y + 6z = 8$$

(CLO-1, GA-2, C2, 7.5 MARKS)

THE END



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI
MID-SEMESTER EXAMINATION OF 2ND SEMESTER 1ST YEAR OF BATCH-
M25/FALL BS ARTIFICIAL INTELLIGENCE OF SPRING 2026

SUBJECT: OBJECT ORIENTED PROGRAMMING (THEORY)

DATED: 25/03/2026

TIME ALLOWED 90 MINUTES

MAX. MARKS 30

Student Name: _____

Student Id Number: _____

Instructions

1. This paper contains 04 questions.
2. Use of Calculator is allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

[Marks 10]

Q.NO.1 (CLO-1 C2 GA2)

- a. EXPLAIN the concept of encapsulation in object-oriented programming. Why is it important?
- b. EXPLAIN the purpose of constructors and destructors in C++.
- c. DISCUSS the different types of constructors.
- d. ILLUSTRATE the class below in the form of UML diagram.

```
class Channel {  
private:  
    string name;  
    int subscribers;
```

```
protected:  
    string creator;
```

```
public:  
    Channel(string n, string c);  
    void displayInfo();  
    void subscribe();  
};
```

Q.NO.2 (CLO-2 C3 GA3)

[Marks 10]

- a. A school keeps track of students and the courses they enroll in. Currently, the system uses a procedural approach. Student information is stored in arrays for names and IDs, and course information is stored in arrays for course names and IDs. Functions like addStudent(name, id) and addCourse(name, id) are used to add students and courses. To enroll a student in a course, the function enrollStudent(studentID, courseID) links the student to the course. Finally, displayCourses() lists all available courses. While this works, it separates data from operations and makes it harder to model real-world relationships, such as a student taking multiple courses. **CHANGE** this procedural system into an object-oriented design by identifying the classes, their attributes, and member functions.
- b. USE the code given below to briefly state the responses of questions associated with it.


```

class Counter
{
private:
    int count;

public:
    Counter();
    Counter(int);
    Counter operator ++ ();
    void display();
};

```

```

main()
{
    Counter c1(4), c2;

    c2 = ++c1;

    c1.display();
    c2.display();
}

```

- How many member functions does class Counter have?
- How many constructors are defined in class Counter?
- Write the definition of the default constructor of class Counter, defined outside of class, so that count is initialized to 0.
- Write the definition of the operator++() function so that it increments the value of count.
- Predict the output produced when display() is called for both objects.

Q.NO.3 (CLO-1 C2 GA2)

[Marks 05]

PREDICT the outputs of the C++ code snippets given below:

```

a      #include <iostream>
        #include <stdlib.h>

        using namespace std;

        class Student
        {
        private:
            static int Total_Students;

        public:
            Student() {

                Total_Students++;
            }

            int How_Many()
            {
                return Total_Students;
            }

```

```

~Student()
{
    Total_Students--;
}
};

```

```

int Student::Total_Students = 0;

```

```

int main()
{
    Student Ali, Sara, Ahmed;
    cout << Ali.How_Many() << " Students are Present" << endl;
    Student *Fatima = new Student;
    cout << Fatima->How_Many() << " Students are Present" << endl;
    delete Fatima;
    cout << Sara.How_Many() << " Students are Present" << endl;
    system("PAUSE");
    return 0;
}

```

b

```

#include <iostream>
using namespace std;

```

```

class A
{
public:
    A()
    {
        cout << "Constructor of A\n";
    }
};

```

```

class B : public A
{
public:
    B()
    {
        cout << "Constructor of B\n";
    }
};

```

```

int main()
{
    B obj;
}

```

c

```

#include <iostream>
using namespace std;

```

```

class Box {
    int length;
public:
    Box() { length = 5; }
}

```



```
void showLength() { cout << length << endl; }  
};
```

```
int main() {  
    Box b;  
    b.showLength();  
    return 0;  
}  
#include <iostream>  
using namespace std;
```

```
class Box {  
    int length;  
public:  
    Box(int l) { length = l; }  
    Box operator+(Box b) { return Box(length + b.length); }  
    void showLength() { cout << length << endl; }  
};
```

```
int main() {  
    Box b1(10), b2(20);  
    Box b3 = b1 + b2;  
    b3.showLength();  
    return 0;  
}
```

```
#include <iostream>  
using namespace std;
```

```
class Distance {  
    int feet;  
    float inches;  
public:  
    Distance() { feet = 0; inches = 0; }  
    Distance(int f, float i) { feet = f; inches = i; }
```

```
void addDistance(Distance d) {  
    inches += d.inches;  
    feet += d.feet + int(inches) / 12; // convert every 12 inches to 1 foot  
    inches = int(inches) % 12 + (inches - int(inches));  
}
```

```
void show() {  
    cout << feet << " feet " << inches << " inches" << endl;  
}  
};
```

```
int main() {  
    Distance d1(5, 9.5), d2(3, 4.7);  
    d1.addDistance(d2);  
    d1.show();  
  
    Distance d3;  
    d3.addDistance(d1);
```

```
d3.show();
```

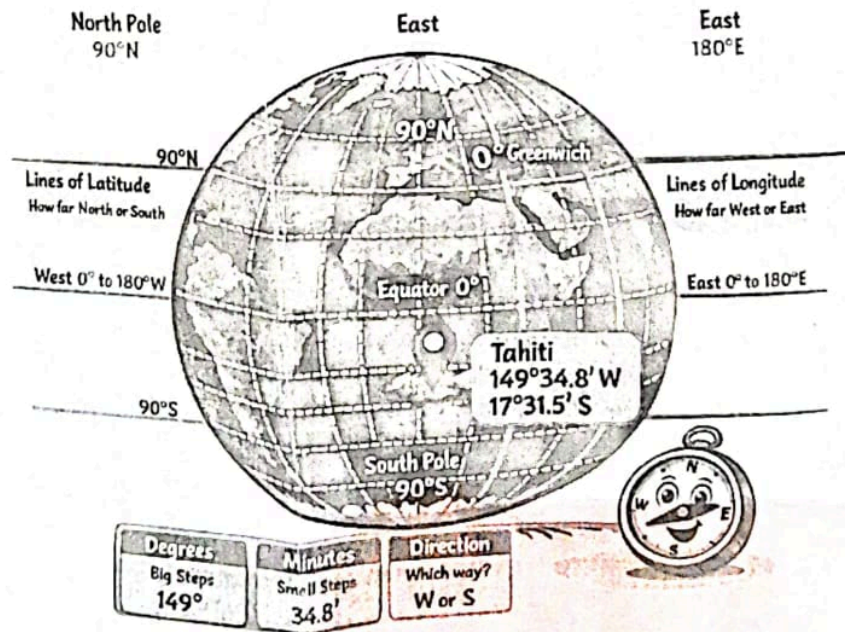
```
return 0;
```

```
}
```

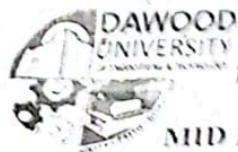
Q.NO. 4 (CLO-3 C3 GA4)

[Marks 05]

In ocean navigation, locations are measured in degrees and minutes of latitude and longitude. Thus if you're lying off the mouth of Papeete Harbor in Tahiti, your location is 149 degrees 34.8 minutes west longitude, and 17 degrees 31.5 minutes south latitude written as 149°34.8' W, 17°31.5' S. There are 60 minutes in a degree. (An older system also divided a minute into 60 seconds, but the modern approach is to use decimal minutes instead.) Longitude is measured from 0 to 180 degrees, east or west from Greenwich, England, to the international dateline in the Pacific. Latitude is measured from 0 to 90 degrees, north or south from the equator to the poles. **APPLY** your knowledge for the given scenario to create a class angle that includes three member variables: an int for degrees, a float for minutes, and a char for the direction letter (N, S, E, or W). This class can hold either a latitude variable or a longitude variable. Write one member function to obtain an angle value (in degrees and minutes) and a direction from the user, and a second to display the angle value in 179°59.9' E format. Also write a three-argument constructor. Write a main() program that displays an angle initialized with the constructor, and then, within a loop, allows the user to input any angle value, and then displays the value. You can use the hex character constant '\xF8', which prints a degree (°) symbol.



THE END



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

Name of Subject Teacher: Faisal

MID EXAMINATION OF 2ND SEMESTER, 1ST YEAR (FALL SEMESTER 2025) OF BS
ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2, & A3), OF 2026

SUBJECT: (PAKISTAN STUDIES)

DATED: 27-03-2026

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name: _____

Student Id Number: _____

Instructions:

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(CLO-1, PLO-6, 7 MARKS)

Q.NO. 1 (A) The Indus Valley Civilization belongs to the **Bronze Age** and is regarded as one of the most advanced urban civilizations of the ancient world due to its remarkable achievements in **architecture, town planning, drainage systems, and urban organization**. Discuss the main characteristics of the Indus Valley Civilization and elaborate the four phases into which historians have divided its development.

(CLO-1, PLO-6, 8 MARKS)

Q.NO. 2 (A) Sher Shah Suri was not only a brave military commander but also one of the most capable administrators in the history of the Indian subcontinent. In the light of the above statement, discuss the major administrative reforms introduced by Sher Shah Suri and evaluate their impact on the effective governance and stability of the Sur Empire.

THE END